

# Statistics 151a: Linear models

UC Berkeley, Spring 2026

Table 1: Calendar (tentative)

| Date   | Day       | Note                   | Unit                           | Topic                                | Assignment   |
|--------|-----------|------------------------|--------------------------------|--------------------------------------|--------------|
| Jan 21 | Wednesday | Lecture                | Introduction                   | Class policies                       | Lab 0        |
| Jan 23 | Friday    | Lecture                |                                | Real-world questions                 |              |
| Jan 26 | Monday    | Lecture                |                                | Inference and probability            |              |
| Jan 28 | Wednesday | Lecture                |                                | Prediction and uncertainty           | Lab 1        |
| Jan 30 | Friday    | Lecture                |                                | Linear algebra review                | HW 0         |
| Feb 2  | Monday    | Lecture                | Unit 1: Multilinear regression | Simple linear regression             |              |
| Feb 4  | Wednesday | Lecture                |                                | Multilinear regression               | Lab (review) |
| Feb 6  | Friday    | Lecture                |                                | ————                                 | Quiz 0       |
| Feb 9  | Monday    | Lecture                |                                | Paths to the least squares objective |              |
| Feb 11 | Wednesday | Lecture                |                                | Transforming the regressors          | Lab 2        |
| Feb 13 | Friday    | Lecture                |                                | Transforming the response            |              |
| Feb 16 | Monday    | Administrative holiday |                                |                                      |              |
| Feb 18 | Wednesday | Lecture                | Unit 2: Testing and inference  | Confidence intervals review          | Lab 3        |
| Feb 20 | Friday    | Lecture                |                                | The multivariate normal              | HW 1         |
| Feb 23 | Monday    | Lecture                |                                | Z and T tests                        |              |
| Feb 25 | Wednesday | Lecture                |                                | Model selection and F-tests          | Lab (review) |

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| Feb 27 | Friday    | Lecture      |                                             | —                                    | Quiz 1       |
| Mar 2  | Monday    | Lecture      |                                             | Asymptotics and consistency          |              |
| Mar 4  | Wednesday | Lecture      |                                             | Heteroskedasticity and grouping      | Lab 4        |
| Mar 6  | Friday    | Lecture      |                                             | The bootstrap                        |              |
| Mar 9  | Monday    | Lecture      | Unit 3: Regularization and machine learning | Splines and basis expansions         |              |
| Mar 11 | Wednesday | Lecture      |                                             | Ridge regression                     | Lab 5        |
| Mar 13 | Friday    | Lecture      |                                             | Lasso regression                     | HW 2         |
| Mar 16 | Monday    | Lecture      |                                             | Cross validation and model selection |              |
| Mar 18 | Wednesday | Lecture      |                                             | Information criteria (AIC and BIC)   | Lab (review) |
| Mar 20 | Friday    | Lecture      |                                             | —                                    | Quiz 2       |
| Mar 23 | Monday    | Spring break |                                             | —                                    |              |
| Mar 24 | Tuesday   | Spring break |                                             | —                                    |              |
| Mar 25 | Wednesday | Spring break |                                             | —                                    |              |
| Mar 26 | Thursday  | Spring break |                                             | —                                    |              |
| Mar 27 | Friday    | Spring break |                                             | —                                    |              |
| Mar 30 | Monday    | Lecture      | Unit 4: How regression can go wrong         | Outliers and leverage                |              |
| Apr 1  | Wednesday | Lecture      |                                             | The influence function               | Lab 6        |
| Apr 3  | Friday    | Lecture      |                                             | Regression to the mean               | HW 3         |
| Apr 6  | Monday    | Lecture      |                                             | The FWL theorem                      |              |
| Apr 8  | Wednesday | Lecture      |                                             | Omitted variable bias                | Lab (review) |
| Apr 10 | Friday    | Lecture      |                                             | —                                    | Quiz 3       |
| Apr 13 | Monday    | Lecture      | Unit 5: Generalizations                     | Logistic regression                  |              |
| Apr 15 | Wednesday | Lecture      |                                             | Poisson regression                   | Lab 7        |
| Apr 17 | Friday    | Lecture      |                                             | Nonlinear least squares              | HW 4         |

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| Apr 20 | Monday    | Lecture |                       | Random effects models |                |
| Apr 22 | Wednesday | Lecture |                       | Hierarchical modeling | Lab (review)   |
| Apr 24 | Friday    | Lecture |                       | Hierarchical modeling | Quiz 4         |
| Apr 27 | Monday    | Lecture | Unit 6: Special topic | TBD                   |                |
| Apr 29 | Wednesday | Lecture |                       | TBD                   | Lab (projects) |
| May 1  | Friday    | Lecture |                       | TBD                   | HW 5           |
| May 4  | Monday    | Lecture | Final projects        |                       |                |
| May 6  | Wednesday | Lecture |                       |                       | Lab (projects) |
| May 8  | Friday    | Lecture |                       |                       |                |

Unless otherwise noted, the primary materials for the course are the lecture notes, which will be posted to the course website in advance of class. The following textbooks are useful supplementary texts and are all either freely available online or available electronically through the UC Berkeley library:

- [SMT: Statistical Models: Theory and Practice](#) Freedman
- [LME: Linear Models and Extensions](#) Ding
- [ISL: An Introduction to Statistical Learning](#) James, Witten, Hastie, Tibshirani
- [VDS: Veridical Data Science](#) Yu, Barter
- [ETM: Econometric Theory and Methods](#) Davidson, MacKinnon

Other books of interest to the class are

- [FPP: Statistics](#) Freedman, Pisani, Purves
- [RDS: R for Data Science](#), Wickham, Grolemund
- [ROS: Regression and other Stories](#) Gelman, Hill, Vehtari